

Sahithi Maram

Senior Software Development Engineer | Distributed Systems | AWS & Cloud Infrastructure

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SUMMARY

I build the backend systems you never see but always depend on. From 6K to 50K TPS at Amazon, I've scaled infrastructure that keeps fulfillment centers running without missing a beat.

SKILLS

Languages & Technologies: C++, Java, Python, JavaScript, TypeScript, Go, SQL, NoSQL, HTML, Linux, React, Node.js, REST APIs

Distributed Systems: Large-scale infrastructure, Cloud storage systems, Data consistency, High availability, Microservices architecture, TCP/IP, Event-driven architecture

Cloud & Tools: AWS (DynamoDB, Lambda, CloudWatch, AppConfig, CDK, CloudFormation, S3, EC2, SQS), Amazon Coral, Guice, Git, Docker

Engineering Practices: System Design (HLD & LLD), Performance tuning at scale, CI/CD pipelines, TDD, Observability, Algorithms & Data Structures, Object-Oriented Design, Design Patterns, Agile/Scrum

AI-Assisted Development: AWS Foundations of Prompt Engineering (Certified), Code generation tooling (Claude, Amazon CodeWhisperer), LLM-assisted development workflows

PROFESSIONAL EXPERIENCE

Amazon

08/2022 – Present | Bellevue, WA

Software Development Engineer

- Tech Lead for distributed systems projects: Drove Dimensions Platform, Inbound Physical Buffer Tracking, and Physical-Virtual Mismatch Mitigation, improving planning accuracy, scalability, and data integrity across 20+ fulfillment centers
- Launched Dimensions Platform processing 1M+ dimension records daily, improving flow planning accuracy and boosting throughput by 35% across fulfillment operations
- Designed scalable multi-type dimension model with optimized caching strategies and data structures, enabling real-time decision-making for distributed fulfillment systems
- Architected distributed data lifecycle management system using DynamoDB TTL and AWS Lambda, processing millions of stale records with automated cleanup
- Implemented idempotent retry logic, fault-tolerant eventual consistency patterns, and automated recovery - eliminating 100K+ invalid entries and improving dashboard accuracy by 40%
- Deployed distributed buffer tracking system across 15+ fulfillment sites, processing 10M+ inventory events daily with sub-second latency and horizontally scalable architecture handling 5K+ TPS
- Built idempotent ingestion pipeline with DynamoDB and Lambda, implementing TTL-based data lifecycle management enabling real-time workforce allocation insights
- Co-led migration to Full Continuous Deployment with staged rings, automated canaries, quality gates, and auto-rollback, significantly increasing release velocity
- Designed dual-track beta/main CI/CD pipelines, streamlining peak-season operations and enabling fast emergency patches, saving ~8 weeks of development time annually
- Led performance optimization of Tier-1 distributed services, scaling system from 6K to 50K TPS while maintaining p99 latencies through caching strategies and query optimizations
- Executed large-scale load tests simulating realistic traffic patterns, identified data access layer bottlenecks, and enhanced observability with structured logging and metrics
- Mentored engineers and interns, improved onboarding processes, and elevated team-wide standards for code quality, testing, and observability

Wipro Technologies

09/2019 – 08/2021 | Hyderabad, India

Software Engineer

- Developed and maintained features for Ericsson's networking tools platform, improving frontend responsiveness and enhancing CI pipeline reliability
- Partnered with QA to design and implement automated test suites, strengthening release hardening and reducing production regressions
- Designed and implemented telecom networking protocol handlers in C++ and Java, focusing on performance optimization and memory management for large-scale distributed network systems handling enterprise traffic
- Collaborated with cross-functional teams in an Agile environment, improving product stability and accelerating delivery cycles

EDUCATION

University of Dayton

08/2021 – 08/2022 | Dayton, OH

Master's in Computer Science | GPA: 3.74/4.00

Bhoj Reddy Engineering College for Women

08/2015 – 05/2019 | Hyderabad, India

Bachelor of Technology in Computer Science

WORK AUTHORIZATION: H1B visa holder